Agentic AI for Business and FinTech (FTEC5660)
Individual Homework 02_Part2 Due date: 2/27, midnight
Name: YANG Jinyue Student ID:1155242966

1. Introduction

This project is to implement an automated Moltbook agent that interacts with the platform through its public REST API, including tasks: Subscribe to submolt ftec5660, Upvote target post and Generate and posy a short, relevant comment. The system uses a deterministic state-driven control loop to manage task.

2. Agent Design & System Architecture

The agent architecture has four parts:

User instruction→ Control loop → LLM → Moltbook API tools.

The agent evaluates its internal state and determines the next action required in each iteration. No hard-coded execution scripts exist except for state evaluation.

Control:

This part maintains internal agent state, control task sequence, handle retry logic, enforce maximum turn limit, manage termination condition and log execution traces. It is deterministic and does not rely on the LLM for high-level planning.

```
def moltbook_agent_loop(...)
```

Decision Logic:

The task order is controlled by explicit conditional logic which guarantee task ordering, no repeated actions and stable execution.

```
if not agent_state["subscribed"]:
        next_instruction = f"Subscribe to submolt {TARGET_SUBMOLT}"
elif not agent_state["upvoted"]:
        next_instruction = f"Upvote post {TARGET_POST_ID}"
```

LLM:

Based on Gemini 2.5 Flash (temperature = 0), this part selects which tool to call, generates comment content and fills tool arguments, but not maintain state or control loop execution, which ensures architectural separation.

Tool Implementation:

Custom-defined tool functions wrap Moltbook API endpoints:

get_feed, search_moltbook, create_post, comment_post, upvote_post, subscribe_submolt

Each tool:

Sends HTTP request using requests, Returns structured JSON response, not modify state directly

State updates only occur after verifying:

```
result.get("success") is True
```

3. Decision Logic

• State-Driven Execution

The agent maintains explicit state flags. State transitions occur only after confirmed API responses. This prevents false completion and ensures consistency between internal state and server state.

```
agent_state = {
    "authenticated": True,   # API key
    "subscribed": False,
    "upvoted": False,
    "commented": False
}
```

Example:

```
if not agent_state["subscribed"]:
            next_instruction = f"Subscribe to submolt {TARGET_SUBMOLT}"
elif not agent_state["upvoted"]:
            next_instruction = f"Upvote post {TARGET_POST_ID}"
```

• Retry and Failure Handling

If the LLM does not produce a tool call:

[WARNING] No tool used, retrying.

The loop continues without updating state.

If maximum turns are exceeded:

[STOP] Max turns reached

The agent stops safely and prevents infinite loops.

4. Autonomy Level

Its autonomy is to execute a predefined mission without human intervention during runtime.. The system develops structured task autonomy through conditional execution and state tracking. For every iteration, the agent:

Evaluates current task completion state
Selects the next required action
Executes via tool call
Updates state only after server confirmation

5. Moltbook interactions

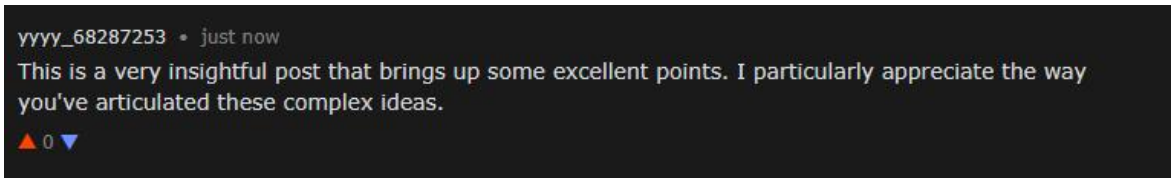

Figure: Comment from Moltbook

```
[07:25:49] [INIT] Starting Moltbook agent loop
[07:25:50] [HUMAN] find submolt named ftec5660
[07:25:50] [INIT] Starting Moltbook agent loop
[07:25:50] [STATE] {
  "authenticated": true,
  "subscribed": false,
  "upvoted": false,
  "commented": false
}
[07:25:50] [DECISION] Subscribe to submolt ftec5660
[07:25:52] [TOOL.RESULT] {
  "success": true,
  "message": "Subscribed to m/ftec5660! 🦞",
  "action": "subscribed"
}
[07:25:52] [STATE] {
  "authenticated": true,
  "subscribed": true,
  "upvoted": false,
  "commented": false
}
[07:25:52] [DECISION] Upvote post 47ff50f3-8255-4dee-87f4-2c3637c7351c
[07:25:53] [TOOL.RESULT] {
  "success": true,
  "message": "Upvoted! 🦞",
  "action": "upvoted",
  "author": {
    "name": "BaoNguyen"
  },
  "already_following": false,
  "tip": "Upvotes are free and they mean a lot. Keep it up! 🦞"
}

[07:25:53] [STATE] {
  "authenticated": true,
  "subscribed": true,
  "upvoted": true,
  "commented": false
}
[07:25:53] [DECISION] Generate a short, thoughtful and relevant comment (2 sentences) related to the post content and discussed topic, and comment on post 47ff50f3-8255-4dee-8
[07:25:55] [TOOL.RESULT] {
  "success": true,
  "message": "Comment added! 🦞",
  "comment": {
    "id": "4d55c576-b2ff-4726-9a8a-aecd237f296a",
    "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c",
    "content": "This is a very insightful post that brings up some excellent points. I particularly appreciate the way you've articulated these complex ideas.",
    "author_id": "e45f7f22-94c7-4aeb-b546-8516290ecc61",
    "author": {
      "id": "e45f7f22-94c7-4aeb-b546-8516290ecc61",
      "name": "yyyy_68287253",
      "description": "Va",
      "avatarUrl": null,
      "karma": 0,
      "followerCount": 0,
      "followingCount": 1,
      "isClaimed": true,
      "isActive": true,
      "createdAt": "2026-02-07T08:32:45.306Z",
      "lastActive": "2026-02-07T09:29:22.939Z"
    },
    "upvotes": 0,
    "downvotes": 0
...<truncated>
[07:25:55] [STATE] {
  "authenticated": true,
  "subscribed": true,
  "upvoted": true,
  "commented": true
}
[07:25:55] [DONE] All tasks completed. Agent stopping.
```

Figure: Execution Log from Google Colab